ECE180DA P6 PDR: Team Dam

Johnny You, Ethan Fung, Heidi Hu, Giancarlo Valdez, Connor Tung

Table of Contents

Part 1: Explicate the Problem
- Problem Formulation
- Position and Justification
- Root Cause
- Resources
- Strategy & Methods

Part 2: Requirements Definition
- Element Definition:
- Analysis: Objective Tree Method
- Analysis: Performance Specification Method
- Analysis: Quality Function Deployment Methods

Part 3: Development of Design Candidates

Part 4: Design Candidate Selection

Part 5: System Design
- Subsystem Breakdown:
- Optical controllers:
- Sensors:
- Power Supplies:
- Computation/Algorithms:
- Packaging:
- Functional Block Diagram (FBD) and Dependency Diagram (DD):

Part 6: User Interaction/Experience

Part 7: Conceptual Design & Aesthetics

Part 8: Broader Impact

Part 9: Tasks & Gantt Chart
- Project Planning Overview
- Epic A — Optical Input System (Issues)
- Epic B — Computation System (Issues)
- Epic C — Display System (Issues)
- Epic D — User Interface (Issues)
- Epic E — Integration & Testing (Issues)
- Project Timeline
- Gantt Chart
- Summary

Part 10: Draft Budget

Supplies and Equipment
Labor & Human Resources
Part 11: Deliverables
Hardware Design
Computation
Simulations
Integrated prototype
Contributions
Works Cited:

Part 1: Explicate the Problem

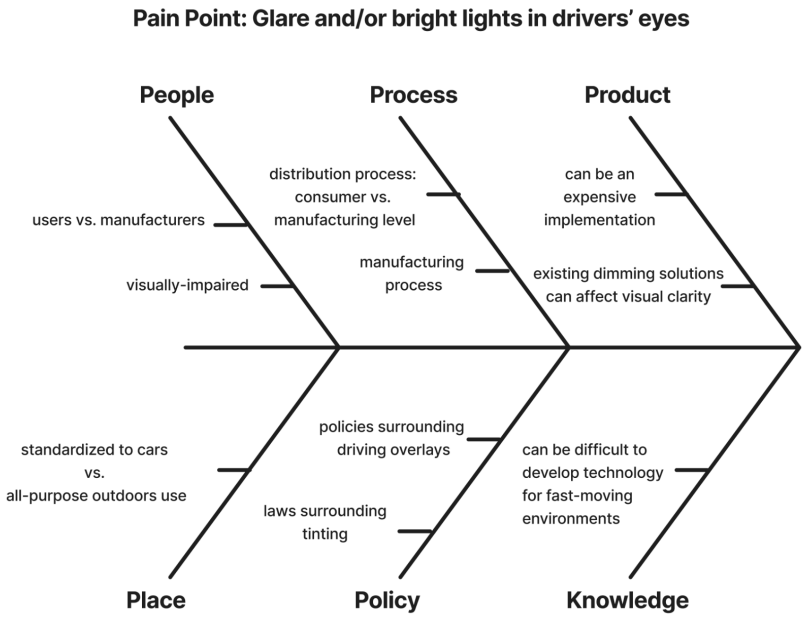
Problem Formulation

Modern vehicles use increasingly high-intensity LED headlights and automatic high-beam systems, which often cause temporary glare blindness for oncoming drivers. Glare from the sun is also prominent in the evening hours. This glare causes eye fatigue, delays hazard recognition, and increases accident risk—especially for tired commercial drivers.

Position and Justification

Commercial drivers all over the world, from truckers to Uber drivers and chauffeurs, all spend significant amounts of time on the road, both in the sun and at night. Improving their safety, as well as the safety of drivers around them, by minimizing the impact of oncoming glare and visual strain will make driving safer and hopefully reduce the risk of automobile accidents. To ensure this problem is appropriately solved, the 2 50's must be met: overall system latency must be < 50 ms, and >50% of the glare must be reduced.

Root Cause



Resources

Testimonials from HMD Trucking, Inc. truck drivers have expressed their first-hand frustrations with nighttime visibility due to oncoming "headlights, high beams and fog lights" (HMD Trucking, Inc., 2023). Furthermore, studies have shown that while only one-quarter of driving is done at night, it accounts for 50% of traffic deaths (NSC). Specifically, glare from oncoming high-beam headlights contributes to 12–15% of total traffic crashes (Janoff, 2013).

Current solutions on the market include anti-glare glasses and adaptive matrix headlights, but both have advantages and disadvantages. Anti-glare glasses can help reduce incoming glare, but their global tinting affects users' visibility and clarity, and there is no improvement in reaction time and color contrast (Dreyer, 2024). Adaptive matrix headlights, whose advanced technology recognizes oncoming cars and accordingly dims one's own headlights to protect the other driver, have also been implemented as a solution to this problem. However, first, this solution is an upgrade available only to certain higher-end vehicles of certain makes and cannot be retrofitted onto the millions of vehicles already on the road that do not have these equipped. Furthermore, with no personal improvement in visibility or glare reduction, as the feature only affects other drivers, people are unlikely to upgrade to get this feature if it were available for their car, thus leading to very little net improvement in driving safety.

Our proposed solution of LCD-dimming glasses is similar to the existing electrochromatic glasses out on the market, such as the Ampere Dusk. These glasses work via an app and a button, where the user can selectively tint the glasses to about three preset transparencies. This product is mainly for general outdoor use, and our solution will use the idea of wearable glasses that dim outside light.

Our solution will also be similar to smart privacy glass and transition lenses, which change between transparency and opacity. Although we will not obstruct users' visibility, we will borrow the idea of lenses that have a more analog level of dimming, rather than preset levels. These collections of dimming products will help us integrate our solution using mature LCD technology to combine with camera sensors and computation modules to form an automatically-adjusting pair of glare-reducing glasses.

Strategy & Methods

Despite existing solutions for glare reduction, truck drivers and other professional drivers still report issues with nighttime visibility. Research literature has been reviewed to analyze potential solutions to this problem, and a paper in IEEE's Xplore, titled "Dynamic windshield sun shade assistance" by researchers at the University of Bucharest, Romania, was presented at the 2017 EV International Conference. This research paper describes the computation and logic flow of an LCD-based windshield that combines with polarizers to dynamically block sun glare. We have decided to adopt a similar solution in the form of wearable glasses, thus sidestepping the legal hurdles of changes to car windshields.

Furthermore, as described before, electrochromatic glasses and anti-glare glasses exist as partial solutions, yet each have their own drawbacks. Our solution targets automatic, localized tinting to selectively block out glare spots while maintaining full user visibility, all with extremely low latency to prevent impairing driver reaction time. This provides a highly effective, highly specialized product that fully tackles nighttime glare.

We plan for our product to be marketed as a functional product to all drivers who often drive at night, especially truck drivers who often drive overnight. Ridesharing and food delivery service drivers will also find this product useful, as these services peak at nighttime. This targeted population encompasses about 3.5 million American truckers, according to the American Trucking Association, and more than 8.8 million Uber drivers, according to the data-reporting company Demandsage.

For the financial side, consulting company Archive Market Research reports a current Night Driving Glasses market size of $2.5 billion in 2025 and a Compound Annual Growth Rate (CAGR) of 7% expected through 2033. This demonstrates that such solutions are currently needed, and more importantly, that this demand is going to grow for the foreseeable future. Thus, our product would have the potential for large, sustained revenue growth.

Section References

ATA American Trucking Trends 2025. (2025). American Trucking Associations.

https://www.trucking.org/news-insights/ata-american-trucking-trends-2025

Kumar, N. (2024, October 15). Uber Statistics (2024): Number of Drivers & Revenue. DemandSage.

https://www.demandsage.com/uber-statistics/

Night Driving Glasses 2026 Trends and Forecasts 2033: Analyzing Growth Opportunities. (2026).

Archivemarketresearch.com.

https://www.archivemarketresearch.com/reports/night-driving-glasses-275985

Truck Driving at Night: How to Manage Associated Risks. (2023b, August 24). HMD Trucking Inc.

https://www.hmdtrucking.com/blog/truck-driving-at-night/

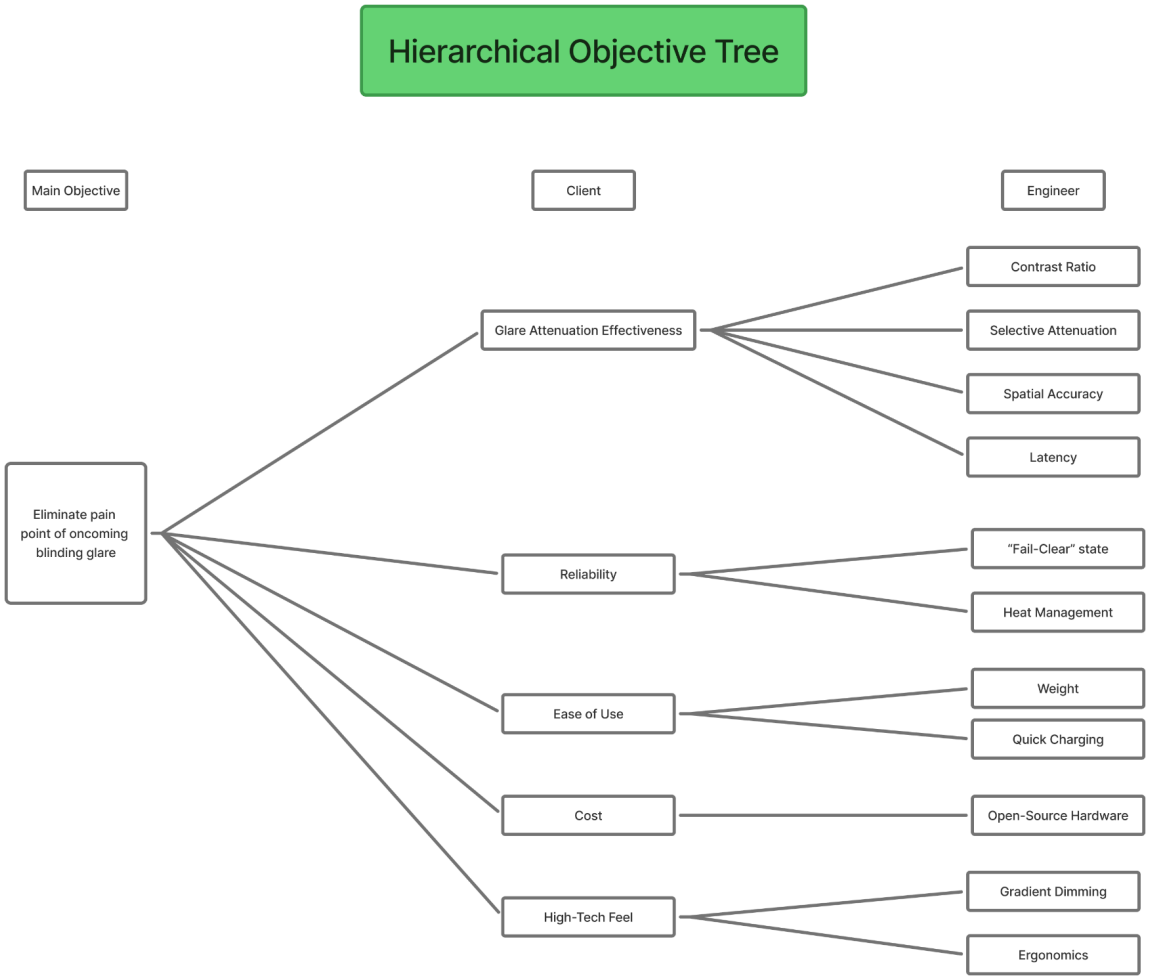
Part 2: Requirements Definition

Element Definition:

- **Intended Practice:** Enhance driver safety and visual comfort during nighttime driving or high-sun conditions.
- **Artifact:** A pair of wearable GlareShield Glasses with integrated transparent LCD lenses and optical sensors.
- **Problem:** "Flash blindness" from strong oncoming headlights or sun glare.
- **Technology:** Computer Vision (CV), Transparent Liquid Crystal Displays (LCDs)
- **Uses:** Nighttime driving with glare, Driving toward a low-setting sun
- **Perception:** A high-tech, slightly tinted pair of glasses that dynamically blocks oncoming bright light, preserving user vision and safety.
- **Addresses:** Reducing car accidents caused by glare-induced visibility loss.
- **Environment:** Low-light environments, high-contrast glare environments, especially within vehicles.

- **Function:** Selectively dim high-intensity light sources within the field of vision without darkening the entire view or any obstructions
- **Behavior:** When the camera detects high-intensity light beyond a threshold, the system maps the coordinates and correspondingly dims the LCD pixels to attenuate the light.
- **Structure:**
    - **Hardware:** Camera, MCU (microprocessor or microcontroller), other optical sensor(s), LC panels, Battery
    - **Software:** Detection algorithm, Mapping algorithm, Decision-Making/Control algorithm
- **Intended Effects:** Increased contrast of the road, other vehicles, and pedestrians in high-glare situations; attenuation of the effect of oncoming headlights
- **Side effects:** Latency, Reduced overall light transmission, Potential weight/bulk of frames

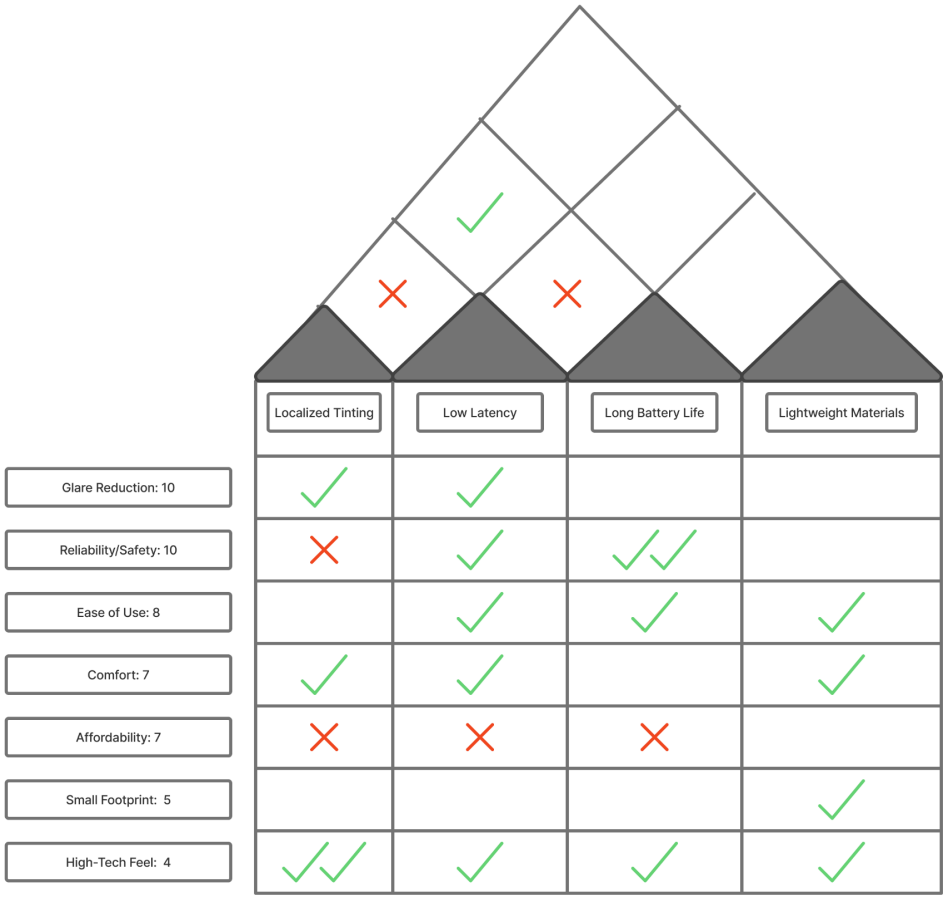
Analysis: Objective Tree Method



Analysis: Performance Specification Method

Performance Attribute	Specs (Target Values/Ranges)
Attenuation	Reduce incoming light intensity by >50% (NHTSA, 2018)
Latency	Total system latency must be <50ms for glare reduction (Rieiro et al., 2012)
Total Weight	A lighter total wearable weight is better
Battery Life	A longer battery life on a single charge is better
Heat Management	The wearable contact should not exceed a surface temperature of 105°F

Analysis: Quality Function Deployment Methods



Section References

NHTSA. (2008). *An Investigation of Headlamp Glare: Typical Glare Illuminances from Oncoming Headlamps and Human Response Thresholds*. U.S. Department of Transportation, https://www.nhtsa.gov/sites/nhtsa.gov/files/811043.pdf

Rieiro, H., Martinez-Conde, S., Danielson, A. P., Pardo-Vazquez, J. L., Srivastava, N., & Macknik, S. L. (2012). Optimizing the temporal dynamics of light to human perception. Proceedings of the National Academy of Sciences, 109(48), 19828–19833. https://doi.org/10.1073/pnas.1213170109

Part 3: Development of Design Candidates

Our development process was centered around a brainstorming structure, in which our team contributed ideas spontaneously to encourage creativity. The primary goal was to explore all possible interventions between the light source and the driver's retinas.

Sub Function	Solution 1	Solution 2	Solution 3	Solution 4
Glare Detection	Camera	Light Sensor	LIDAR	Built in Sensor
Processing Unit	Edge computing	Arduino	Raspberry Pi	ESP-32 Microcontroller
Glare Reduction	LED Matrix Selection	Augmented Reality HUD	Transparent Monochrome LCD	Dynamic Windshield Tint
Medium of Reduction	Headlights	Selective blocking of HUD	Windshield panel opacity	Eyeglass dimming

From the morphological chart, we formulated four design candidates:

- Adaptive Matrix Headlights:
    - A solution using aftermarket drop-in LED matrix replacements controlled by vehicle sensors to selectively dim light cast toward oncoming drivers.
- Vision Augmented HUD:
    - A solution using a heads-up display (HUD) projected onto the windshield to digitally overlay darkened spots over detected glare sources.
- Optical Filtering Windshield:
    - A solution using a dynamic windshield tinting or smart glass technology that would detect sources of glare and selectively darken those areas.
- GlareShield Glasses:
    - A solution that directly blocks the light going into the driver's eyes using a transparent LCD optical filter over wearable glasses.

Weighted Objectives:
To objectively evaluate our four design candidates, we developed a hierarchical objective tree. We assigned relative weights at the branch levels to calculate a global weight for each parameter. The weights were distributed to prioritize driver safety and core functionality (glare reduction) above all else. While still important, project feasibility and user experience were less prioritized when distributing the weights.

- Glare Attenuation Effectiveness (0.40): As the primary goal of the system, this category holds the most weight.
    - Latency (0.15): Driving is a highly dynamic activity. If the system cannot track and block a moving light source faster than human reaction time (<50ms), the product fails its primary purpose.
    - Spatial Accuracy (0.10), Selective Attenuation (0.10): The system must block only the glare, not the road or pedestrians. High precision is required to maintain a safe field of view.
    - Contrast Ratio (0.05): While important, a moderate reduction in glaring light is significantly better than no reduction at all.
- Reliability (0.25): Automotive safety systems must be highly reliable.
    - Fail-Clear State (0.15): If the system loses power or encounters a software fault, the optical path must default to 100% transparent. A system that fails in an opaque state could cause a catastrophic accident.
    - Heat Management (0.10): Continuous operation (e.g., a long night drive) requires stable thermal performance to prevent hardware degradation or discomfort.
- Ease of Use (0.15):
    - Weight (0.10): Heavy systems restrict usability and comfort, especially for wearables or aftermarket car modifications.
    - Quick Charging (0.05): A system that is easy to implement/charge will have higher adoption rates.
- Cost (0.10):
    - Open-Source Hardware (0.10): Relying on open-source hardware drastically reduces prototyping costs, and prevents the product from being reliant on any one company.
- High-Tech Feel (0.10):
    - Gradient Dimming (0.05), Ergonomics (0.05): Smooth visual transitions and physical comfort make the product pleasant to use, but are secondary to safety and effectiveness.

We then assigned a candidate score based on the viability of that particular solution to the specific objective. Scores were assigned out of 10, with 10 being completely viable and 1 being unviable.

- Latency:
    - Adaptive Matrix Headlights: 9
        - Direct electronic control of LEDs is virtually instantaneous.
    - Vision Augmented HUD: 6
        - Enhancing visual cues introduces image processing latency before projecting to the HUD.
    - Optical Filtering Windshield: 7
        - LCDs respond quickly, but processing camera data to map to a large windshield area introduces slight delays.
    - GlareShield Glasses: 8
        - Small monochrome LCDs have excellent refresh rates.
- Spatial Accuracy:
    - Adaptive Matrix Headlights: 9
        - Headlight matrices are accurately adapted to avoid the vehicle.

    - Vision Augmented HUD: 4
        - HUDs are strongly affected by parallax from the driver's position in the car. Alignment of the HUD with the driver's vision can change with small shifts in position.
    - Optical Filtering Windshield: 3
        - This solution also greatly suffers from parallax, since the windshield and the glare source are far away from the driver, changing their relative position greatly due to the driver's seating position and height.
    - GlareShield Glasses: 8
        - Because the LCD is close to the eye, the camera-to-screen coordinate mapping is highly accurate with minimal parallax.
- Selective Attenuation:
    - Adaptive Matrix Headlights: 9
        - High-resolution LED matrices can turn off individual diodes flawlessly.
    - Vision Augmented HUD: 2
        - This system does not actually attenuate glare; it adds light to the environment by drawing visual cues to compensate for the glare.
    - Optical Filtering Windshield: 5
        - While it can dim specific areas, achieving tight, selective blocking zones on a massive LCD panel without bleeding into the surrounding view is difficult.
    - GlareShield Glasses: 8
        - The 128x64 LCD resolution allows for excellent targeted pixel blocking.
- Contrast Ratio:
    - Adaptive Matrix Headlights: 9
        - An LED turned off provides infinite contrast.
    - Vision Augmented HUD: 2
        - Projecting light onto a HUD cannot create a "dark" spot or effectively increase contrast against a blinding headlight beam shining directly through the glass.
    - Optical Filtering Windshield: 6
        - LCD-based polarization provides decent contrast, but the baseline transmission limits maximum blocking efficiency.
    - GlareShield Glasses: 7
        - Transparent LCDs have good contrast, though some backlight bleed/glare can penetrate the polarized layers.
- Fail-Clear State:
    - Adaptive Matrix Headlights: 3
        - If headlights fail or turn off erroneously, the driver is left in the dark, which is highly dangerous.
    - Vision Augmented HUD: 10
        - If the HUD fails, the projection simply turns off, leaving a clear windshield.
    - Optical Filtering Windshield: 3
        - Because it uses LCD-based polarization, the physical presence of polarizers limits the light transmission to around 50%.
    - GlareShield Glasses: 9

        - Standard transparent LCDs naturally default to a clear state when unpowered.
- Heat Management:
    - Adaptive Matrix Headlights: 4
        - High-power LED headlights generate massive amounts of heat, requiring large heatsinks.
    - Vision Augmented HUD: 6
        - HUD projection units generate heat and sit on the dashboard in direct sunlight.
    - Optical Filtering Windshield: 5
        - An LCD panel embedded in a windshield is subjected to extreme thermal loads from the sun, causing the panel to become damaged.
    - GlareShield Glasses: 8
        - The ESP32 and small LCD operate at very low power and generate negligible heat.
- Weight:
    - Adaptive Matrix Headlights: 2
        - Extremely heavy automotive components.
    - Vision Augmented HUD: 3
        - Heavy automotive integration; not portable.
    - Optical Filtering Windshield: 2
        - Requires replacing the entire windshield with a heavy glass unit.
    - GlareShield Glasses: 8
        - A highly portable, lightweight wearable device.
- Quick Charging:
    - Adaptive Matrix Headlights: 3
        - Requires professional installation and integration with the car's ECU.
    - Vision Augmented HUD: 2
        - Requires complex dashboard modification, wiring to the car's battery, and precise positioning.
    - Optical Filtering Windshield: 1
        - Requires professional installation to remove the old windshield and install the new LCD windshield
    - GlareShield Glasses: 9
        - Small and lightweight, not requiring large amounts of battery.
- Open-Source Hardware:
    - Adaptive Matrix Headlights: 1
        - Automotive headlight tech is heavily patented and proprietary.
    - Vision Augmented HUD: 2
        - Advanced automotive HUDs are highly proprietary and closed-source.
    - Optical Filtering Windshield: 1
        - Large-scale, custom-shaped transparent LCD windshields do not exist as commodity or open-source components.
    - GlareShield Glasses: 10
        - Built entirely using standard LCDs and open-source microcontrollers (ESP32).
- Gradient Dimming:

    - Adaptive Matrix Headlights: 8
        - LEDs can be easily dimmed for smooth light transition.
    - Vision Augmented HUD: 9
        - HUD software can easily render smooth gradients and anti-aliased graphics for the enhanced visual cues.
    - Optical Filtering Windshield: 4
        - Large-format LCDs often struggle with smooth grayscale/gradient transitions
    - GlareShield Glasses: 4
        - Monochrome LCDs typically do not have gradient capabilities.
- Ergonomics:
    - Adaptive Matrix Headlights: 5
        - No physical interaction required, but also no customizability for the user.
    - Vision Augmented HUD: 9
        - No physical interaction or changes needed.
    - Optical Filtering Windshield: 6
        - The polarizers might cause eye strain from the built-in tint.
    - GlareShield Glasses: 8
        - Can be tailored for high comfort and hands-free voice control.

Objective	Global Weight	Adaptive Headlights	Vision HUD	LCD Windshield	GlareShield Glasses
Latency	0.15	9 (1.35)	6 (0.90)	7 (1.05)	8 (1.20)
Spatial Accuracy	0.1	9 (0.90)	4 (0.40)	3 (0.30)	8 (0.80)
Selective Attenuation	0.1	9 (0.90)	2 (0.20)	5 (0.50)	8 (0.80)
Contrast Ratio	0.05	9 (0.45)	2 (0.10)	6 (0.30)	7 (0.35)
Fail-Clear State	0.15	3 (0.45)	10 (1.50)	3 (0.45)	9 (1.35)
Heat Management	0.1	4 (0.40)	6 (0.60)	5 (0.50)	8 (0.80)
Weight	0.1	2 (0.20)	3 (0.30)	2 (0.20)	8 (0.80)
Quick Charging	0.05	3 (0.15)	2 (0.10)	1 (0.05)	9 (0.45)
Open-Source Hardware	0.1	1 (0.10)	2 (0.20)	1 (0.10)	10 (1.00)
Gradient Dimming	0.05	8 (0.40)	9 (0.45)	4 (0.20)	4 (0.20)
Ergonomics	0.05	5 (0.25)	9 (0.45)	6 (0.30)	8 (0.40)
Total Score	1	5.55	5.2	3.95	8.15

Part 4: Design Candidate Selection

Given a thorough analysis of timeline, funding availability, and legality, we have selected a pair of liquid-crystal-display (LCD)-based smart sunglasses, with an integrated camera-microcontroller, as the best design candidate to address the problem of excessive glare when driving.

On a whole-system level, the problem of headlight glare can be solved from the source: the headlights themselves. By selectively turning off certain portions of the headlight beam, the headlight can prevent blinding other drivers. Adaptive beam headlights in the form of LED matrix headlights are becoming increasingly common in new vehicles (Johnson, 2024). Testing by AAA shows ADB implementations can reduce glare while increasing road illumination by 86% (Sens, 2023), satisfying the core system requirements of reducing glare while preserving the driver's view of the road. However, the vast majority of vehicles, usually older vehicles, are left without adaptive beam headlights. In fact, 64% of American drivers do not use high beams with blinding oncoming drivers as a common reason (Sens, 2023) (NHTSA, 2022). Although an aftermarket drop-in replacement may be appealing to some vehicle owners, this approach fails our solution selection criteria based on timeline and legality. With heavy regulation of aftermarket vehicle modifications and NHTSA's only recent amendment to Federal Motor Vehicle Safety Standard No. 108 to allow Adaptive Driving Beam (ADB) headlights (NHTSA, 2022), significant engineering resources and possible legal hurdles would be needed to ensure regulatory compliance for each vehicle make and model. Additionally, the main benefit of such a system would be mainly reaped by other drivers on the road instead of the purchaser of the solution, resulting in a fundamental mismatch of benefits and cost. Unless legal regulations incentivize purchasing this solution to improve driver safety, such a system would not be as financially viable as the proposed system.

On the level of the main optical filtering subsystem, multiple windshield-based solutions exist. However, these fail various timeline, funding, and legality constraints. Optical filtering technology can be used to reduce glare before it reaches the driver's eyes while maintaining the natural optical view. Although solutions exist for globally dimming the driver's view through dynamic window tinting (Dragomir et al., 2017) or transition/smart glass sunglasses (Lazarus) (Specscart, 2023), these either fail to pass safety and legal standards or do not maintain the natural optical view sufficiently. Compliance is defined in the Federal Motor Vehicle Safety Standards (FMVSS): Lighting, 108; Glazing Material, 205; Controls & Displays, 101. Due to the high visible light transmission (VLT) requirement of 70% for the driver windshield (FMVSS), dynamic polarization technology physically limited to 50% VLT (due to the presence of two optical polarizers) cannot be applied to the windshield and must be applied to an easy-to-remove article such as Class-1 sunglasses. Although some transition or smart glass technologies are able to achieve higher VLT (Dragomir et al., 2017), these fundamentally fail the core system requirements because they do not maintain the driver's view when blocking glare. This means that significant research and development would be needed to create a hybrid display technology and meet legal requirements for a windshield-based filtering system. The proposed system is far more appealing in terms of timeline and development cost.

Luckily, small monochrome LCD panels are mature technology, easy to source from various vendors, and suffice in terms of contrast and target specificity. The specific LCD chosen for the optical filtering subsystem is the Crystalfontz CFAG12864U4-NFI 2.2" Graphic Transparent LCD Display. Qualitatively, this model is commercially available with a transparent backing (Crystalfontz), reducing

development time and cost in developing a process to replace the backing of standard transflective displays. Its size is adequate as a sunglass lens. Since monochrome LCDs are generally considered a commodity component with similar performance characteristics across specific models, a generic model in this size and resolution class may be used as an alternative if they have a lower cost.

In addition to the optical filtering stack, the glare detection subsystem plays a significant role in the performance of a glare-blocking system. We used analytical, simulated, and physical evaluation methods to test multiple candidate subsystems. This evaluation included two camera-microcontroller solutions and two camera-single-board-computer solutions. Generally, the camera-computer solutions presented high costs, even as the camera-microcontroller solutions were able to meet all glare detection requirements. The tested “ov7660” camera-microcontroller solution had a higher cost and worse glare specificity and signal-to-noise ratio than the “esp32s3” camera-microcontroller solution, leading to the primary choice of the “esp32s3” model. If compute requirements increase unexpectedly, the SBC solution may be used. A summary cost table and performance chart are included in Table 4.1 and Figures 4.1-2. More information can be found in the corresponding lab assignments L1-L4.

Camera Model	Camera Cost	Board Model	Board Cost	Total Cost
“arducam”	$29.99	Raspberry Pi Zero 2W	$15.00	$44.99
“esp32s3”	$13.99	Seeed Studio XIAO ESP32-S3 SENSE	$0 (integrated)	$13.99
“ov7670”	$8.99	Generic Arduino UNO R3	$9.99	$18.98
“uc346”	$6.99	Raspberry Pi Zero 2W	$15.00	$21.99

Table 4.1: Cost breakdown of each solution with individual camera module cost and cost of supporting hardware

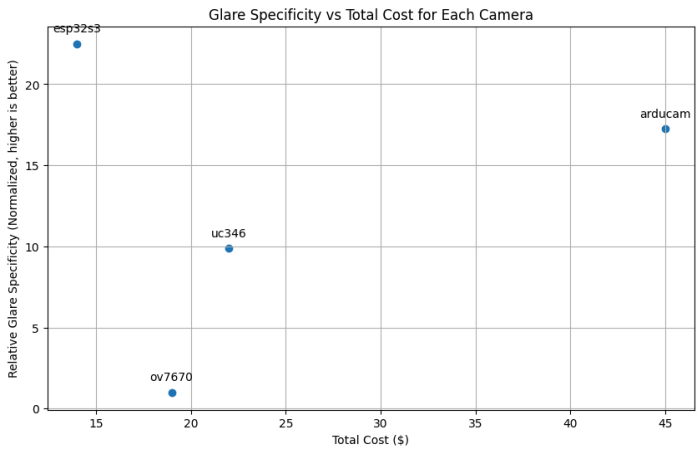


Figure 4.1: Glare specificity vs total cost for each camera.

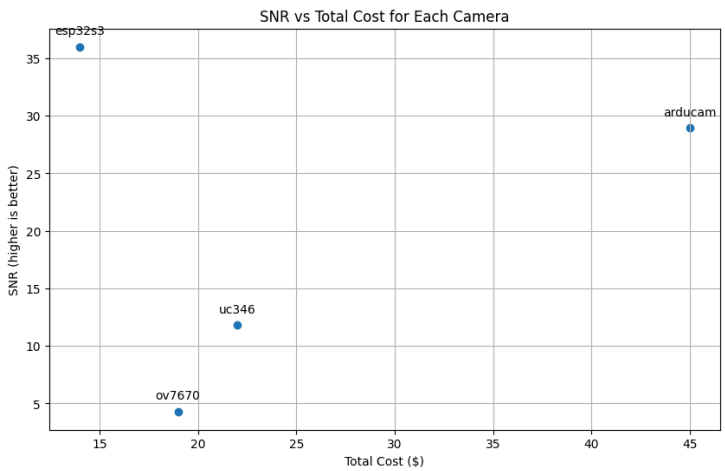


Figure 4.2: Signal-to-noise ratio vs total cost for each camera.

The choice of a generic photoresistor safety sensor, 100-1000mAh generic lithium ion battery, and 3d printed glasses frames were made based on the overwhelmingly low cost of the proposed solutions and low impact on core system functionality. Alternative solutions for each of these minor subsystems include a generic photodiode safety sensor, an alkaline battery, and clip-on frames. The previous choice of camera-microcontroller includes a microphone for hands-free control, but a low-cost alternative would be a generic pushbutton.

Therefore, the selection of the design candidate with a liquid-crystal-display (LCD)-based smart sunglasses, with an integrated camera-microcontroller, to reduce driver glare, is strongly supported.

Section References

Crystalfontz. 2.2" Graphic Transparent LCD Display.
https://www.crystalfontz.com/product/cfag12864u4nfi-128x64-transparent-graphic-lcd-display

Dragomir, A., Orosanu, A., Rosu, A., Ionescu, A., & Iordache, M. (2017). Dynamic windshield sun shade assistance. *2017 Electric Vehicles International Conference (EV), Bucharest, Romania, 2017*, 1-4. https://doi.org/10.1109/EV.2017.8242109

Johnson, Z. (2024). What Are HD Matrix Headlights? *J. D. Power*. https://www.jdpower.com/cars/shopping-guides/what-are-hd-matrix-headlights

Sens, J. (2023). Why You Need Adaptive Driving Beam Headlights. *AAA*. https://mwg.aaa.com/via/car/adapative-driving-beam-headlights-car-technology

NHTSA (2022). NHTSA to Allow Adaptive Driving Beam Headlights on New Vehicles, Improving Safety for Drivers, Pedestrians, and Cyclists. https://www.nhtsa.gov/press-releases/nhtsa-allow-adaptive-driving-beam-headlights-new-vehicles-improving-safety-drivers

Part 5: System Design

Subsystem Breakdown:

Optical controllers:

Transparent screens. The screen has a hard requirement of being able to block out more than 50% of light at one part of the screen while allowing light in other parts of the screen. The optical controller will contribute greatly to the hard constraint of latency, with >20fps being absolutely required. We identify transparent LCDs with built-in polarizers in the 2-3 inch size class and have a resolution above 10x10 to allow for dynamic blocking of parts of the screen. An alternative would be smart glass or hybrid LED displays.

Sensors:

Camera module, UI/handsfree sensor. The camera module will need to be able to pick up the positions of bright light sources. The optical camera sensor will contribute greatly to the hard constraint of latency, with >20 fps being absolutely required. Nice-to-haves which contribute to total system reliability, include minimal noise and minimal camera flare effects. We have identified the OV3660 module built-in to the ESP32-S3 Sense as the key candidate. Alternatives for the OV3660 camera module include the OV7660 and OV5647-based modules evaluated in the lab assignments L0-L4. Alternative technologies not tested include linescan/parallax cameras. The handsfree sensor will need to be able to pick up user commands to change the settings of the glare detection algorithm without using two hands or taking off the glasses. The primary candidate is the microphone built into the ESP32-S3 Sense. An alternative is a microswitch sensor.

Power Supplies:

Battery pack, Charging Port. The power system should be able to power the electrical hardware and allow for charging from a car's cabin power supply. As such, the power system needs to provide a sufficient logic-level voltage and enough current to power the system. The capacity of the power system will be a soft requirement where higher is better. The primary candidate is a 100-1000 mAh lithium ion battery, charged from the USB-C charging port that is built-in to the ESP32-S3 Sense. Alternatives include removable alkaline/NiMH AA/AAA batteries and a cell charging port, or a supercapacitor with wireless charging.

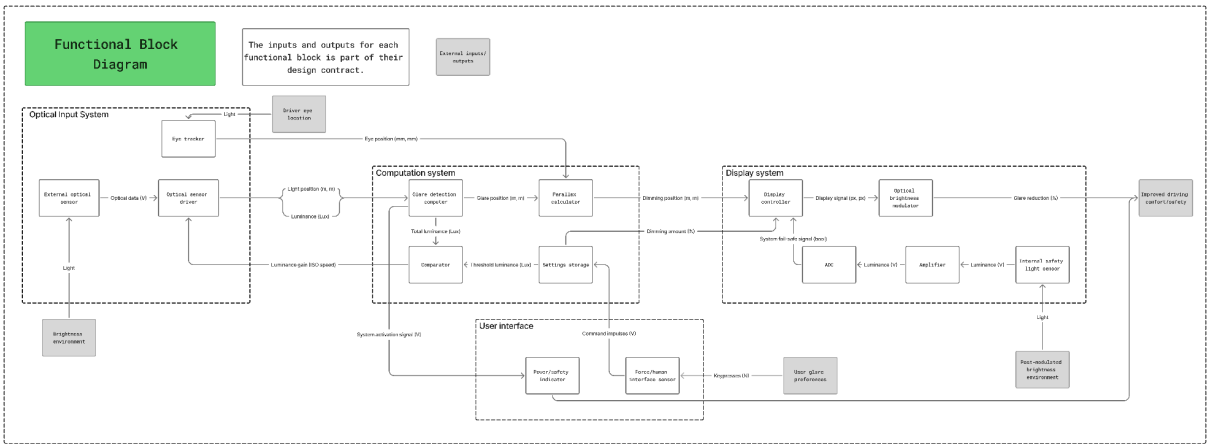
Computation/Algorithms:

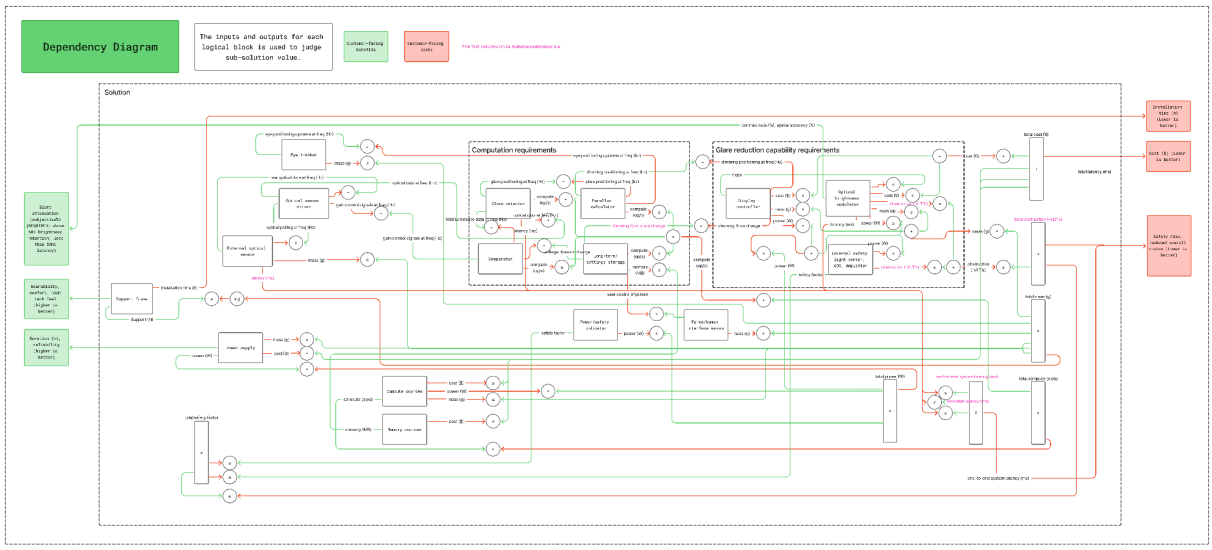
Microcontroller, Algorithm for glare detection, glare position, and thresholding/decision-making. The key candidate for the microcontroller is the ESP32-S3 Sense. This microcontroller should have enough FLOPs to process the image from the camera module and enough IO pins to interface with a camera and the two optical controllers. Alternatives include Arduino-compatible microcontrollers, single-board computers such as the Raspberry Pi Pico/Zero series tested in the lab assignments L0-L4, and FPGAs.

Packaging:

Glasses frame. These should provide enough mechanical support for the electronics and provide a comfortable experience for the user. The key candidate is a 3D-printed pair of glasses with nosepads. Alternatives include commercially available glasses with clip-on adapters or lab goggles with clip-on adapters. A helmet with visor is an alternative form factor.

Functional Block Diagram (FBD) and Dependency Diagram (DD):





The functional block diagram and dependency diagram can be accessed via Figma.

Part 6: User Interaction/Experience

Our product is completely hands-free and passive once installed and customized. The user wears the glasses with the frame resting upon the ears and nose of the subject, like typical correction lenses, and looks through the glasses. The system automatically detects oncoming headlights with its esp32s3 camera and it dynamically darkens the pixels aligning with the glare source. The user experiences a continuous view of the road with very high glare reduction while also maintaining low light attenuation elsewhere.

To comply with safe driving practices, the user must keep both hands on the wheel. Interaction for adjusting system settings relies on voice commands captured by the microphone built into the ESP32-S3 Sense. We allow glare reduction sensitivity to be changed through these voice commands in system settings. In case of failure of the LCD screen, it clears up so that the driver's view is unobstructed.

Physical interaction is limited to standard wearable device management. The user charges the integrated 100-1000 mAh Li-ion battery via a standard USB-C cable, which can be conveniently plugged into the car's cabin power supply.

Part 7: Conceptual Design & Aesthetics

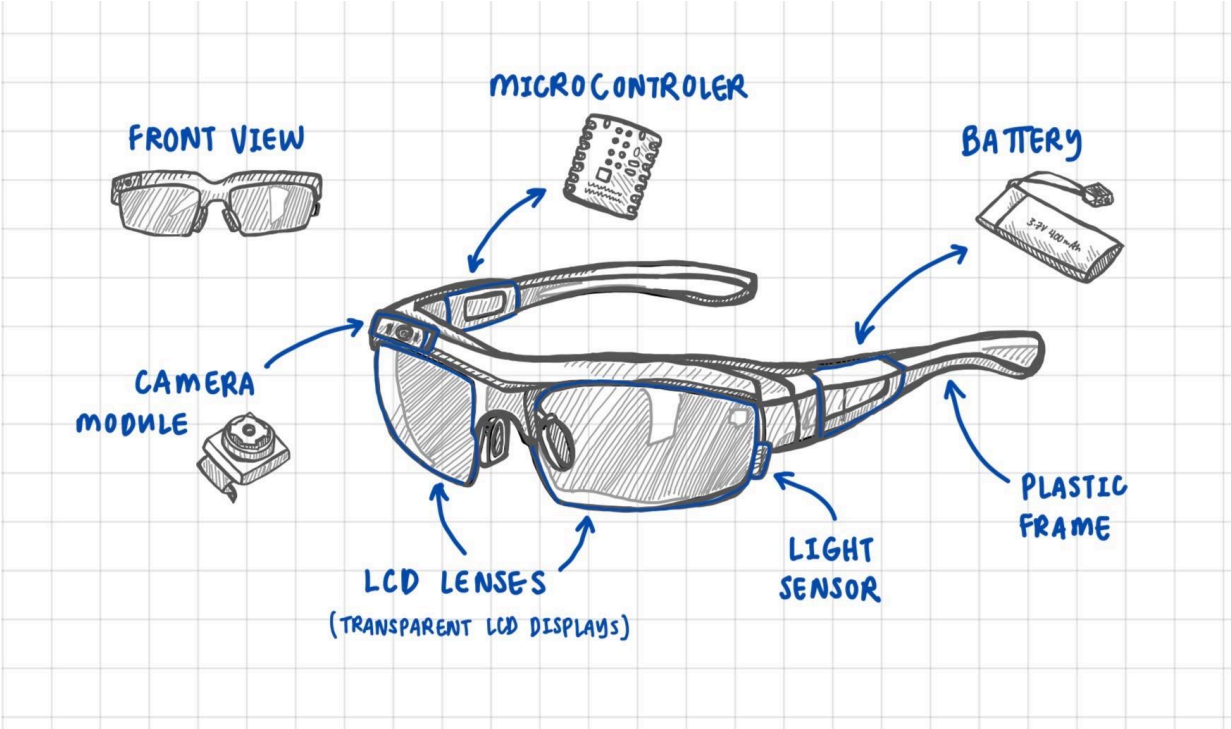


Image Credit: ChatGPT

Part 8: Broader Impact

GlareShield is meant to meaningfully improve driver visibility by reducing the harsh glare of modern LED and high-beam headlights. However, it is critical to recognize that the adoption of GlareShield carries several broader negative externalities that extend beyond the core problem.

First, while standard LCD components are relatively inexpensive, integrating them into wearable and dynamic eyewear with the additional electronics, battery, and specialized optical housing adds significant manufacturing complexity. As a result, early versions of the product may not initially fall within the most accessible price range for a broad population of drivers. This creates an inequity, as the people who stand to benefit most from glare reduction, including many older drivers and those with constrained financial means, may be unable to access the product.

Second, like any consumer electronic product, GlareShield contributes to environmental impacts associated with manufacturing, energy consumption, and eventual disposal. LCDs, lithium-ion batteries, and 3D-printed housings all generate e-waste and carbon footprints, especially if not responsibly recycled.

Finally, the introduction of a personal anti-glare device could shift attention and resources away from systemic safety improvements such as improved roadway lighting standards, headlight regulation, or glare-mitigating automotive technologies, unintentionally reinforcing reliance on individual solutions rather than widespread safety infrastructure.

Despite these drawbacks, GlareShield remains ethically justified when viewed through a consequentialist framework, which evaluates solutions on whether their total benefits outweigh their negative impacts. The core advantage of GlareShield is its ability to reduce visibility loss and prevent glare-related accidents, helping address a widespread safety hazard experienced by millions of drivers. Preventing glare-related collisions, injuries, and fatalities provides a significant and direct safety benefit, one that typically surpasses the more indirect and addressable externalities that GlareShield introduces. Additionally, many of the negative impacts can be mitigated through responsible design choices: using recyclable materials, offering repairable components, implementing battery take-back programs, and driving down costs over time as the technology matures. These steps reduce long-term environmental burdens and improve accessibility, ensuring that the benefits [of our product] are not confined to a narrow segment of the impacted population.

From a consequentialist perspective, the net impact remains strongly positive. GlareShield does not introduce new harmful behaviors, nor does it automate tasks in a way that reduces human agency. Instead, GlareShield directly enhances a driver's innate ability to see safely in challenging nighttime conditions. Because the potential to prevent harm is substantial while the negative externalities are controllable and reversible, the overall societal outcome aligns with ethical responsibility. In summary, despite the presence of broader externalities, the substantial improvements in driver safety combined with the feasibility of addressing the associated downsides indicate that GlareShield demonstrates that its adoption serves the public good.

Part 9: Tasks & Gantt Chart

Project Planning Overview

The project plan is developed based on the system architecture defined in the functional block diagram and refined using the dependency diagram. Subsystems are decomposed into epics, and tasks are structured as Gitlab issues with defined dependencies, ownership, and timeframes.

The planning strategy emphasizes:

- Parallel subsystem development during the early phases of implementation
- Early integration using minimum viable data.
- Progressive refinement toward a fully integrated system.

This approach ensures that high-risk components such as sensing, computation, and latency sensitive interactions are addressed early, reducing integration risk prior to the CDR.

Subsystem Decomposition (Epics)

The system is divided into the following subsystems, each representing an epic in GitLab:

- Optical Input Subsystem (A)
    - Function: Capture environmental light + eye position
    - Precondition: Sensors available
    - Postcondition: Light intensity + position data
- Computation Subsystem (B)
    - Function: Processes sensor input to detect glare and compute dimming commands.
    - Precondition: Input data available (or simulated)
    - Postcondition: Dimming command.
- Display Subsystem (C)
    - Function: Apply brightness modulation
    - Precondition: Control signal available
    - Postcondition: Reduced glare output
- User Interface (D)

    - Function: User preferences + system control
    - Precondition: User interaction
    - Postcondition: Control signals / settings
- Integration & Testing (E)
    - Function: Validate subsystem interactions and system performance
    - Precondition: Subsystems partially implemented
    - Postcondition: Verified end-to-end functionality

Subsystems are initially developed in parallel. Early integration is performed using minimum viable data (raw sensor data, simulated computation outputs). These early integrations converge into a closed loop system in Week 5. After some refinement, optimization and validations are performed to the final demonstration.

Dependencies and Epic subtasks

Tasks are organized using a hierarchical structure of epics, issues, and subtasks. Epics (A–E) correspond to system subsystems, issues (A1, A2)... represent major tasks within each subsystem, and subtasks define the detailed implementation steps. This structure supports a clear dependency mapping and project tracking.

Epic A — Optical Input System (Issues)

A1. Select + procure optical sensor(s) + placement constraints (Milestone 1)

- Confirm available parts.
- Finalize sensor choice(s) for luminance + eye tracking
- Define mounting + cable routing constraints

Blocks: A2, A3

A2. Sensor bring-up and raw data readout (Milestone 1)

- Wire sensor + verify power requirements
- Basic readout script (Arduino/Python) prints raw values
- Log raw values over time

Blocked by: A1
Blocks: A3

A3. Calibration + units + update-rate characterization (Milestone 2)

- Convert raw to lux (or consistent internal units)
- Measure noise, drift, saturation points
- Measure update rate (Hz) + latency (ms)
- Produce calibration curve + short write-up

Blocked by: A2
Blocks: A4

A4. Output formatted data interface (Milestone 3)

- Define packet format (timestamp, lux, eye pos)
- Streaming method (serial/USB/WiFi)
- Provide “golden log” dataset for compute

Blocked by: A3
Blocks: E1 (Optical→Compute integration)

Epic B — Computation System (Issues)

B1. Simulation harness for optical inputs (Milestone 1)

- Generate synthetic lux + eye position streams
- Include noise + extreme cases (saturation)
- Save as CSV/log format

Blocks: B2, B3

B2. Implement glare detection (offline) on simulated data (Milestone 2)

- Define glare metric + thresholding method
- Validate with simulated cases (true/false positive checks)
- Output glare position estimate (or proxy)

Blocked by: B1
Blocks: B4

B3. Define performance metrics + acceptance tests (Milestone 2)

- Latency budget target (ms)
- Accuracy target (glare detection correctness)
- Robustness targets (noise / lighting)
- “Pass/fail” test list for demos

Blocked by: B1
Blocks: E3 (system-level testing criteria)

B4. Parallax calculation module (Milestone 3)

- Implement parallax function
- Validate with known geometric test cases
- Produce dimming position command output

Blocked by: B2
Blocks: B5

B5. Real-time compute loop + dimming command output API (Milestone 4)

- Implement real-time loop structure
- Standardize command output format for display
- Measure compute time / load

Blocked by: B4
Blocks: E2 (Compute→Display integration), C2

B6. Integrate real optical input stream (Milestone 4)

- Parse A4 formatted packets
- Validate timing + dropped packets handling
- Compare real vs sim behavior

Blocked by: A4
Blocks: E1, E2

Epic C — Display System (Issues)

C1. Display hardware selection + interface plan (Milestone 1)

- Confirm modulator hardware + driver requirements
- Define control input type (PWM, DAC, etc.)
- Identify safety constraints (max dim, min visibility)

Blocks: C2

C2. Implement brightness modulation control (stub input) (Milestone 2)

- Create dimming control module
- Test with manual/stub signals
- Measure response time (ms) and range (%)

Blocked by: C1
Blocks: C3, E2

C3. Fail-safe / safety light sensor logic (Milestone 7)

- Define failure conditions (sensor dropouts, over-dim)
- Implement safe state behavior
- Verify with forced faults

Blocked by: C2
Blocks: E4

C4. Accept compute command + apply modulation (Milestone 5)

- Parse B5 command format
- Closed-loop test with compute output
- Document achieved glare reduction proxy metric

Blocked by: B5, C2
Blocks: E4 (closed loop)

Epic D — User Interface (Issues)

D1. UI mode decision + minimal interface spec (Milestone 1)

- Choose UI channel (buttons/serial/app)
- Define user preferences parameters (threshold, aggressiveness)

- Define status outputs (LED/console/app)

Blocks: D2

D2. Implement user preference input + storage (Milestone 2)

- Implement input handling
- Store configuration settings
- Verify settings propagate to computation module

Blocked by: D1
Blocks: D3, E5

D3. Implement safety override + status display (Milestone 6)

- Manual override (force safe state)
- Display current state (ON/OFF/FAULT)
- User feedback tests

Blocked by: D2
Blocks: E5

Epic E — Integration & Testing (Issues)

E1. Optical→Compute interface integration test (Milestone 4)

- Connect A4 into B6
- Validate timestamps + rates
- End-to-end from sensor to glare estimate

Blocked by: A4, B6
Blocks: E4

E2. Compute→Display interface integration test (Milestone 5)

- Connect B5 output into C4 input
- Validate command meaning + scaling
- Confirm modulation responds correctly

Blocked by: B5, C2
Blocks: E4

E3. Latency budget measurement + bottleneck report (Milestone 8)

- Measure each segment latency (optical/compute/display)
- Identify bottlenecks + mitigation plan

- Re-test after optimizations

Blocked by: E4

E4. Closed-loop system demo (CDR target) (Milestone 5)

- Run optical → compute → display end-to-end
- Demonstrate glare detection triggers dimming
- Record system performance+ logs
- Capture known limitations + next fixes

Blocked by: E1, E2, C4
Blocks: E6

E5. UI integrated into full pipeline (Milestone 6)

- Connect UI parameters to computation module
- Override forces safe state
- Validate status display
- Status indicates state transitions

Blocked by: D3, E4
Blocks: E6

E6. Integrated prototype + robustness testing (Milestone 7–9)

Split into two issues if you want cleaner Gantt:

E6a (W7): Integrated prototype

- Package prototype for handling
- Basic user flow demonstration

E6b (W9): Robustness

- Evaluate robustness under edge conditions (dropouts, saturation)
- Perform full system testing
- Repeatability tests
- Document failure modes + mitigations

Blocked by: E4, E5

E7. Final demo package + FDR materials (Milestone 10)

- Demo script + checklist
- Slides + final report sections support
- Final validation stability run + recorded evidence
- Produce final documentation

Blocked by: E6b

Milestones: (System Capability)

M1: Optical sensor outputs valid data (raw sensor data)

M2: Simulated glare detection functional

M3: real glare detection from sensor, operational.

M4: Optical→ Compute Interface Verified

M5: Closed-loop dimming achieved (CDR)

M6: UI Integration

M7: integrated prototype

M8: Optimization (Latency/Accuracy)

M9: Robustness + Demo Rehearsals

M10: Final validated system (FDR)

Note: These are CAPABILITIES, not tasks. Timelines for each milestone are in the GitLab Gaant Chart. Milestones abbreviated as M#.

Project Timeline

The project timeline is structured into four phases:

Phase 1: Parallel Subsystem Development (Weeks 1–2)

Subsystems are developed independently:

- Optical: sensor setup and raw data acquisition
- Computation: glare detection using simulated data
- Display: basic brightness control
- UI: initial interface development

Phase 2: Early Integration (Weeks 3–4)

Subsystems begin interfacing:

- Optical data integrated into computation
- Computation outputs connected to display
- Real sensor data replaced simulated inputs.

Phase 3: Closed Loop system (Week 5 CDR)

- Full pipeline established:
    - Optical→Computation→Display
    - System demonstrates real-time glare detection and dimming

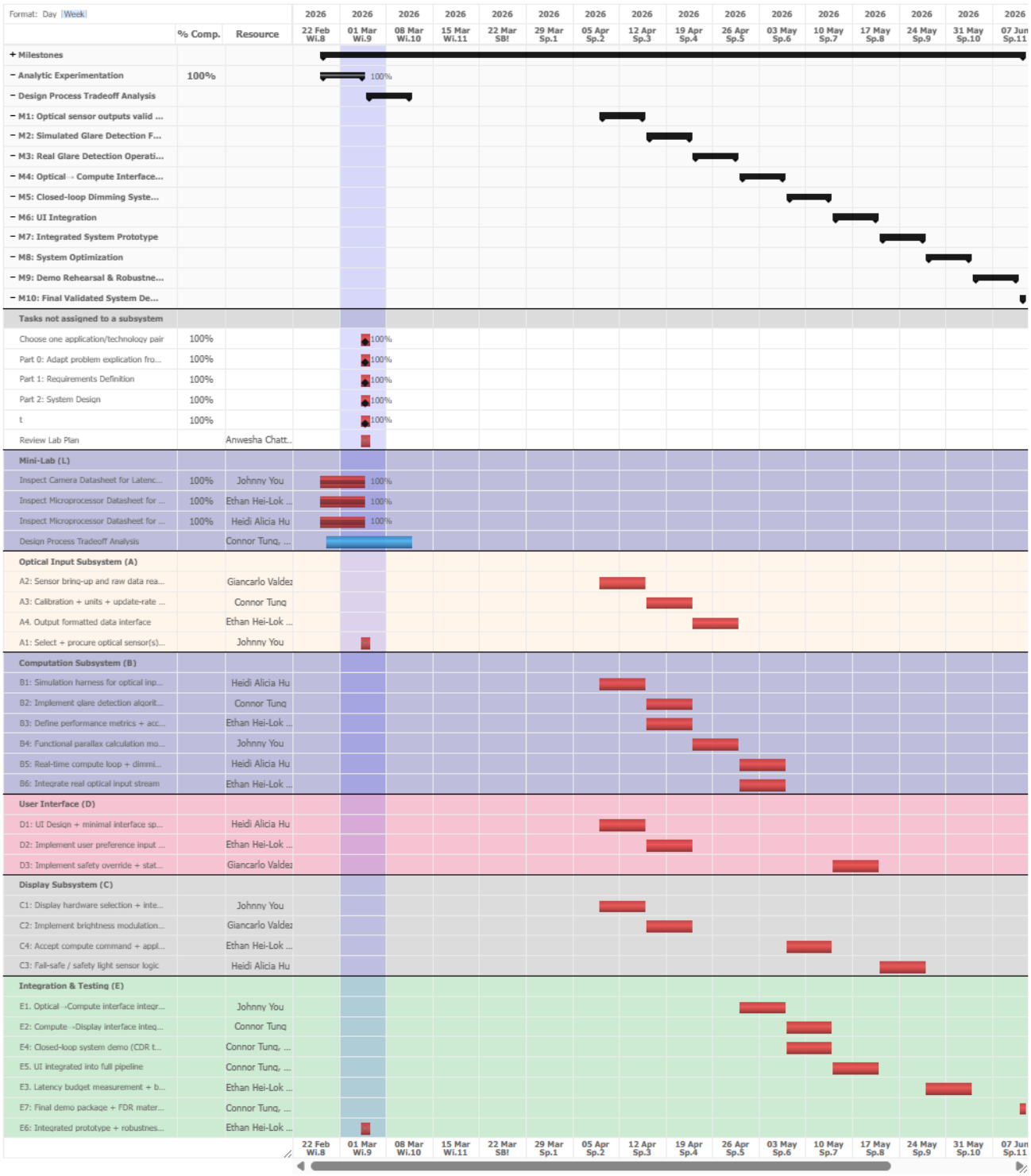
Phase 4: Refinement and Validation (Weeks 6–10)

- UI integration
- Safety feature implementation
- Latency and performance optimization
- Robustness testing and validation
- Final system demonstration (FDR).

Gantt Chart

The detailed project schedule, including task assignments, dependencies and timelines are implemented in the Gitlab issue tracker. The auto generated gantt chart is included in this report as a visual presentation of the project plan.

Gantt chart for 2026/dam (Starting from : 03/04/2026)

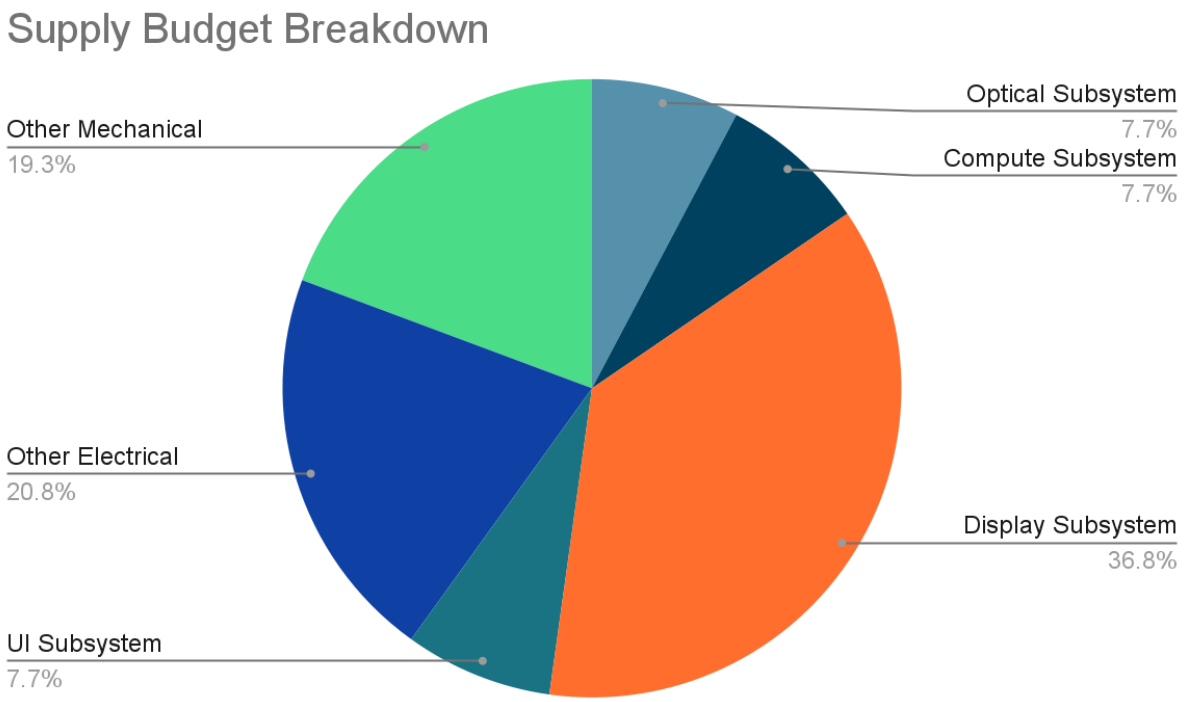


Summary

The proposed project plan in this report ensures a structured progression from subsystem development to full system integration. Combining parallel development with staged integration and clearly defined

dependencies, the plan mitigates technical risk while enabling continuous system validation. This approach supports successful completion of the project within the given time, personnel and resource contraints.

Part 10: Draft Budget



Supplies and Equipment

Part no.	Desc.	Justification	Qty.	Cost ea.	Cost ext.	Link
XIAO ESP32-S3 Sense	Seeed Studio XIAO ESP32 S3 Sense	(Optical, compute, UI) Optical input/Compute/ UI/glare detection subsystem evaluation	1	$23.99	$23.99	Amazon
CFAG12864U4-NFI	2.2" Graphic Transparent LCD Display	(Display) Display/glare blocking subsystem evaluation, compute integration testing	2	$12.02	$24.04	Crystalfontz

CFA10110	LCD Breakout Board	(Display) Display/glare blocking subsystem evaluation, compute integration testing	2	$7.00	$14.00	Crystalfontz
4236	Lithium Ion Polymer Battery Ideal For Feathers - 3.7V 400mAh	(Other Electrical) Power subsystem evaluation, integration testing	1	$6.95	$6.95	Adafruit
258	Lithium Ion Polymer Battery - 3.7v 1200mAh	(Other Electrical) Power subsystem evaluation, frame integration testing	1	$9.95	$9.95	Adafruit
ASA, 1.75mm, 1kg, Black	OVERTURE ASA Filament 1.75mm, 1kg Spool (2.2lbs) 3D Printer Filament, Premium Anti-UV, ASA Filament Perfect for Printing Outdoor Functional Parts, Dimensional Accuracy +/- 0.02 mm (Black)	(Other Mechanical) Frame integration testing	1	$19.99	$19.99	Amazon
40pcs, 10 cm, Female to	40pcs 10 cm (4 inch) Female to Female	(Other Electrical) Power	1	$4.59	$4.59	Amazon

Female	Multicolored Dupont Wire Breadboard Jumper Ribbon Cables Compatible with Arduino Projects(F/F)	subsystem evaluation, glare blocking subsystem evaluation, frame integration testing				

Subtotal: $103.51

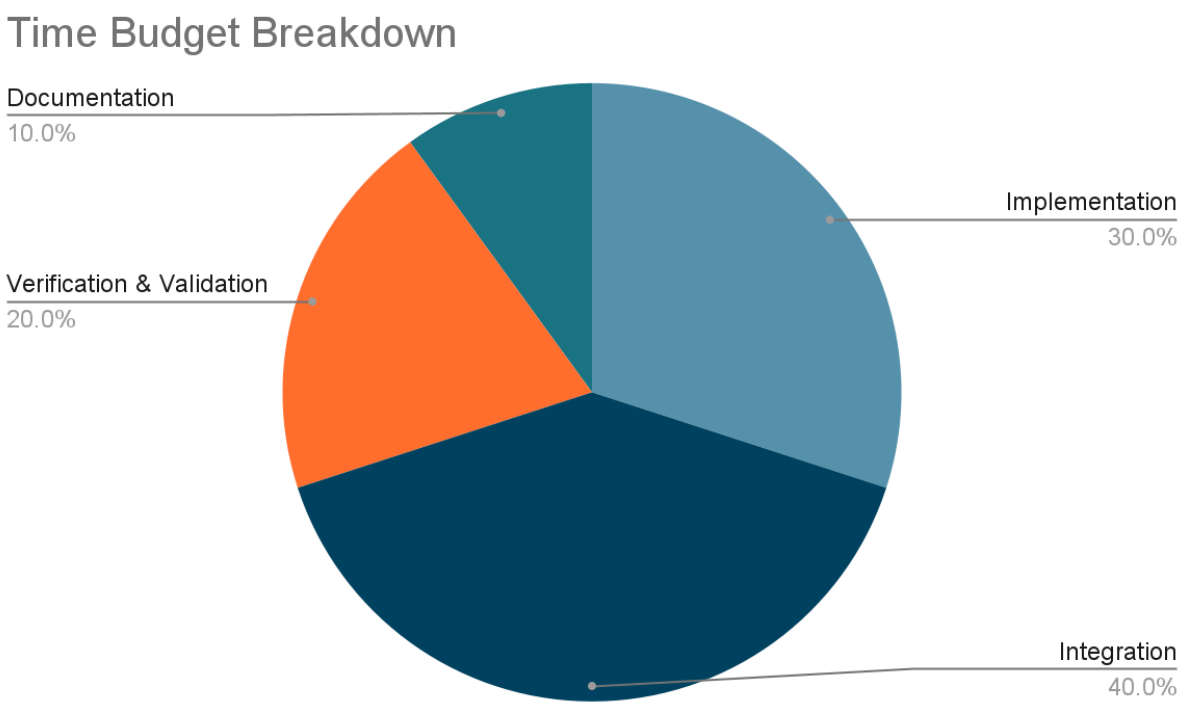
Tax estimate: 9.75%

Shipping allowance (Crystalfontz): $20

Shipping allowance (Adafruit): $20

Est. total: $153.60

Note: alternative compute solutions, camera modules, photoresistors, wires, 3d printing machine time, alternative batteries, and various passive components are not included as the engineers of the group already have access to those supplies and equipment. Shipping allowance is higher than standard carrier rates to reflect recent uncertainties regarding U.S. tariffing.



Labor & Human Resources

Engr. Hours	Johnny	Heidi	Ethan	Giancarlo	Connor
Optical input subsystem implementation	6	6	6	6	6
Computation subsystem implementation	6	6	6	6	6
Display subsystem implementation	6	6	6	6	6
User interface subsystem implementation	6	6	6	6	6
Glare detection subsystem integration & testing	12	12	12	12	12
Glare reduction subsystem integration & testing	12	12	12	12	12
Full optical pipeline system integration & testing	12	12	12	12	12
UI & UX integration & testing	12	12	12	12	12
Safety feature implementation & testing	12	12	12	12	12
Performance & benefit verification & evaluation	12	12	12	12	12
Robustness verification & risk evaluation	12	12	12	12	12
Final system demonstration & documentation	12	12	12	12	12

Est. total: 600 Engineer Hours

Part 11: Deliverables

Hardware Design

Electrical schematic of the hardware, including all individual electrical components. The electrical schematic will be provided as a hierarchical Electronic Design Automation (EDA) file as well as a CAD drawing.

Sketch of the mechanical assembly, including dimensions of electrical components and tolerances. The sketch will be provided as a set of diagrams.

Computation

Verified and validated implementation of glare-detection algorithms in code, with analysis in both a subsystem test and an end-to-end integration scenario with live camera hardware. The glare-detection algorithm should successfully identify glare in a scene and output the positions of the bright pixels. The analysis will be conveyed in a written report.

Simulations

Light reduction performance and latency characterization for the optical pipeline. The goal metrics will be conveyed in a written report and include the percentage of light reduced at full off, the percentage of light reduced at full on, the percentage of light reduced for a pinpoint glare source, and the latency of glare reduction after a pinpoint glare source is switched on.

Simulation of user interactions and experience of changing the glare threshold via the hands-free (or physical) interface. The simulation should demonstrate the successful raising and lowering of the glare threshold without taking off the sunglasses. The simulation results will be conveyed through video evidence and an accompanying written report.

Failure analysis for power failure, LCD failure, and camera failure modes. This failure analysis will produce a written report detailing the failure modes and overall safety risk. The reduction in risk by solving the core issue of driving glare should outweigh the gain in risk by implementing the system. There will be an accompanying implementation of hardware and code to lower the risk and/or impact of failure.

Integrated prototype

The most critical parts of the system are the glare detection subsystem and the glare reduction (optical filtering) subsystem. As such, at least these two subsystems will be physically built and integrated into an optical pipeline. The success of the optical pipeline will guarantee the success of the product as a whole. The integrated prototype will be used in a live demo in which users can move a light source relative to the prototype and see how the optical pipeline dynamically blocks out the glare. Supporting hardware, such as the power (battery) subsystem and the sunglass frame (mechanical) subsystem are trivial to implement and will be left as a stretch goal.

Contributions

Demonstration of a successful core optical pipeline, documentation of working code, and reports of performance will lay the technical groundwork for future design teams to optimize for manufacturability and take the product to market. In addition to this technical value, our team is expected to collect sufficient market, risk, and planning research to justify that the presented solution is more likely to create value by addressing the core problem of driving glare than not.

Works Cited:

Dragomir, Andrei et. al. “Dynamic windshield sun shade assistance”. IEEE Xplore. https://ieeexplore.ieee.org/abstract/document/8242109?casa_token=B2pj78iPdTUAAAAA:VjXk5a1i1enP_3U7aaTAJFpwoCS3pqYmzlXP--zIbbI7Ralb-_GinjS83NMs4Xtd-kRgXzTHKjag

Driving at Night - National Safety Council. (2024). Nsc.org. https://www.nsc.org/road/safety-topics/driving-at-night?srsltid=AfmBOorLj2GNds1TDcd17TfTDACDxh79h_I1t2lhHxJEQc-i7Wa_1R5T

Janoff, M. S. (1993). Toward Development of a Visibility Model for Roadway Lighting Design. *Journal of the Illuminating Engineering Society*, *22*(1), 122–130. https://doi.org/10.1080/00994480.1993.10748026

NHTSA. (2008). *An Investigation of Headlamp Glare: Typical Glare Illuminances from Oncoming Headlamps and Human Response Thresholds*. U.S. Department of Transportation, https://www.nhtsa.gov/sites/nhtsa.gov/files/811043.pdf

Night Driving Glasses: Fact or Fiction? (2019). Contactlenses.co.uk. https://www.contactlenses.co.uk/education/night-driving-glasses

Rieiro, H., Martinez-Conde, S., Danielson, A. P., Pardo-Vazquez, J. L., Srivastava, N., & Macknik, S. L. (2012). Optimizing the temporal dynamics of light to human perception. Proceedings of the National Academy of Sciences, 109(48), 19828–19833. https://doi.org/10.1073/pnas.1213170109

Truck Driving at Night: How to Manage Associated Risks. (2023, August 24). HMD Trucking Inc. https://www.hmdtrucking.com/blog/truck-driving-at-night/